

Table of Contents

1. INTRODUCTION	3
2. FEATURE LIST	4
2.1. System Admin.....	4
2.2. Pension Admin.....	4
2.3. Pension Manager	4
2.4. Viewer.....	5
3. ER DIAGRAM	6
3.1. Normalization	6
3.2. Finalization	8
4. UML DIAGRAM	9
4.1. Use Case Diagram	9
4.2. Activity Diagram	9
5. CONCLUSION.....	15

1. INTRODUCTION

Pension management is an essential task for government institutions, as it directly supports the financial security of retired employees. In many offices, pension records are still maintained manually or through partially digital systems. This often leads to problems such as slow processing, repeated data, lack of transparency, and mistakes in record handling. As a result, managing employee information and tracking pension status becomes difficult and time-consuming.

To address these challenges, this project proposes the development of a Government Pension Management System (Govt. BD Pension). The proposed system will be a desktop-based application developed using C# Windows Forms with database support. Its goal is to create a centralized and secure platform where pension-related employee information can be managed more easily and accurately.

The system will use role-based access control so that different users can access only the features relevant to their roles. Proper data validation and structured record management will help reduce errors and improve efficiency. This project is proposed as an academic application of Object-Oriented Programming (OOP) concepts, GUI design, and database integration, based on a real-life government pension management scenario.

2. FEATURE LIST

The system is designed with role-based functionality, ensuring that each user can access only the features relevant to their responsibilities.

2.1. System Admin

The System Admin holds the highest level of authority in the system. The key features available to this user include:

- Secure login with role-based authentication
- Create, update, and delete system user accounts
- Assign and manage user roles
- Reset user passwords
- Full access to all system modules
- Control over system-level configurations

This role ensures system integrity, security, and proper user management.

2.2. Pension Admin

The Pension Admin is responsible for managing pension-related employee data. The available features include:

- Add new employee records
- Edit and update employee information
- Delete employee records
- Manage pension-related data
- View employee lists in tabular format
- Search employees by ID
- Filter employees by status (Active / Retired)
- Access dashboard analytics

This role focuses on operational management and data accuracy.

2.3. Pension Manager

The Pension Manager role is responsible for monitoring pension-related information without having full administrative control. This role ensures proper supervision while maintaining data integrity.

Features available to the Pension Manager include:

- Secure login with role-based access
- View employee pension details
- Monitor pension status (Active / Retired / Pending)
- View employee service duration and retirement dates
- Access dashboard analytics in read-only mode
- No permission to delete employee records

This role supports verification and monitoring activities within the system.

2.4. Viewer

The Viewer role is designed for users who require limited access to employee information for observation purposes only.

Features available to the Viewer include:

- Secure login with restricted access
- View basic employee information (ID, Name, Department, Status)
- View employee list in read-only mode
- No access to pension calculations or analytics
- No permission to add, edit, or delete any data

This role ensures transparency while protecting sensitive pension information from unauthorized modification.

3. ER DIAGRAM

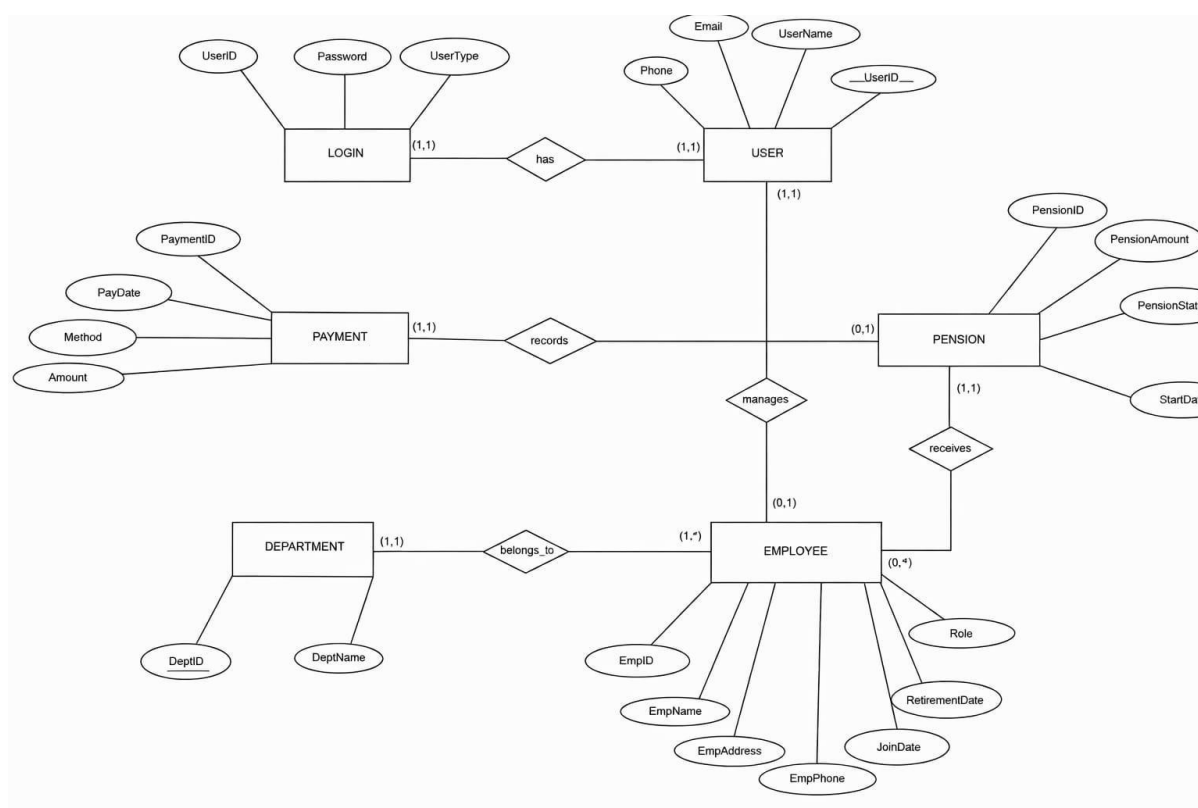


Fig-1 : ER Diagram

The ER Diagram of the Government Pension Management System (Govt. BD Pension) represents the logical structure of the database by showing the main entities, their attributes, and the relationships among them. The diagram is designed to support secure, accurate, and efficient management of pension-related information.

The system consists of six core entities: User, Login, Department, Employee, Pension, and Payment. Each user has a unique login credential, ensuring controlled system access. Employees are assigned to departments, and each employee is associated with a single pension record. Pension records may have multiple payment entries, allowing proper tracking of pension disbursement over time.

Primary and foreign key relationships are used to maintain data integrity and avoid redundancy. The ER diagram reflects real-life pension management workflows and supports role-based access, efficient data retrieval, and scalable database design.

3.1. Normalization

Normalization is used in this project to organize the database structure, reduce data redundancy, and ensure data consistency. The database of the Government Pension Management System (Govt. BD Pension) has been normalized up to Second Normal Form (2NF).

User and Login Relationship

Relationship Type: One-to-One

Each user in the system has a single login account.

Unnormalized Form (UNF):

UserID, UserName, Email, Phone, Password, UserType

First Normal Form (1NF):

The relation is already in 1NF because all attributes contain atomic values.

Second Normal Form (2NF):

To separate authentication information from user profile data, the relation is divided into two tables:

- User (UserID, UserName, Email, Phone)
- Login (UserID, Password, UserType)

Employee and Department Relationship

Relationship Type: One-to-Many

A department can have multiple employees, but each employee belongs to only one department.

Unnormalized Form (UNF):

EmpID, EmpName, EmpAddress, EmpPhone, JoinDate, RetirementDate, Role, DeptID, DeptName

First Normal Form (1NF):

All attributes are atomic, so the table is in 1NF.

Second Normal Form (2NF):

Department information is separated to remove redundancy:

- Employee (EmpID, EmpName, EmpAddress, EmpPhone, JoinDate, RetirementDate, Role, DeptID)
- Department (DeptID, DeptName)

Employee and Pension Relationship

Relationship Type: One-to-One

Each employee is associated with a single pension record.

Unnormalized Form (UNF):

EmpID, EmpName, EmpAddress, EmpPhone, JoinDate, RetirementDate, Role, PensionID, PensionAmount, PensionStatus, StartDate

First Normal Form (1NF):

The table satisfies 1NF as all attributes are atomic.

Second Normal Form (2NF):

Employee and pension information are separated into:

- Employee (EmpID, EmpName, EmpAddress, EmpPhone, JoinDate, RetirementDate, Role)
- Pension (PensionID, PensionAmount, PensionStatus, StartDate, EmpID (*FK*))

Pension and Payment Relationship

Relationship Type: One-to-Many

A pension may receive multiple payments over time.

Unnormalized Form (UNF):

PensionID, PensionAmount, PensionStatus, StartDate, PaymentID, PayDate, Method, Amount

First Normal Form (1NF):

The relation is already in 1NF.

Second Normal Form (2NF):

To avoid repetition of pension details, the relation is divided into:

- Pension (PensionID, PensionAmount, PensionStatus, StartDate)
- Payment (PaymentID, PayDate, Method, Amount, PensionID (*FK*))

Employee Receives Pension

Relationship Type: One-to-One

Each employee receives exactly one pension.

Unnormalized Form (UNF):

EmpID, EmpName, EmpAddress, EmpPhone, JoinDate, RetirementDate, Role, PensionID

First Normal Form (1NF):

The table is already in 1NF.

Second Normal Form (2NF):

The relation is divided into:

- Employee (EmpID, EmpName, EmpAddress, EmpPhone, JoinDate, RetirementDate, Role)
- Pension (PensionID, EmpID (*FK*))

3.2. Finalization

After applying normalization up to Second Normal Form (2NF), the finalized database structure of the Government Pension Management System is:

- Login → UserID, Password, UserType
- User → UserID, UserName, Email, Phone
- Department → DeptID, DeptName
- Employee → EmpID, EmpName, EmpAddress, EmpPhone, JoinDate, RetirementDate, Role, DeptID (*FK*)
- Pension → PensionID, PensionAmount, PensionStatus, StartDate, EmpID (*FK*)
- Payment → PaymentID, PayDate, Method, Amount, PensionID (*FK*)
-

This finalized design ensures proper data organization, minimizes redundancy, and maintains referential integrity, making the system efficient, reliable, and suitable for real-world pension management.

4. UML DIAGRAM

4.1. Use Case Diagram

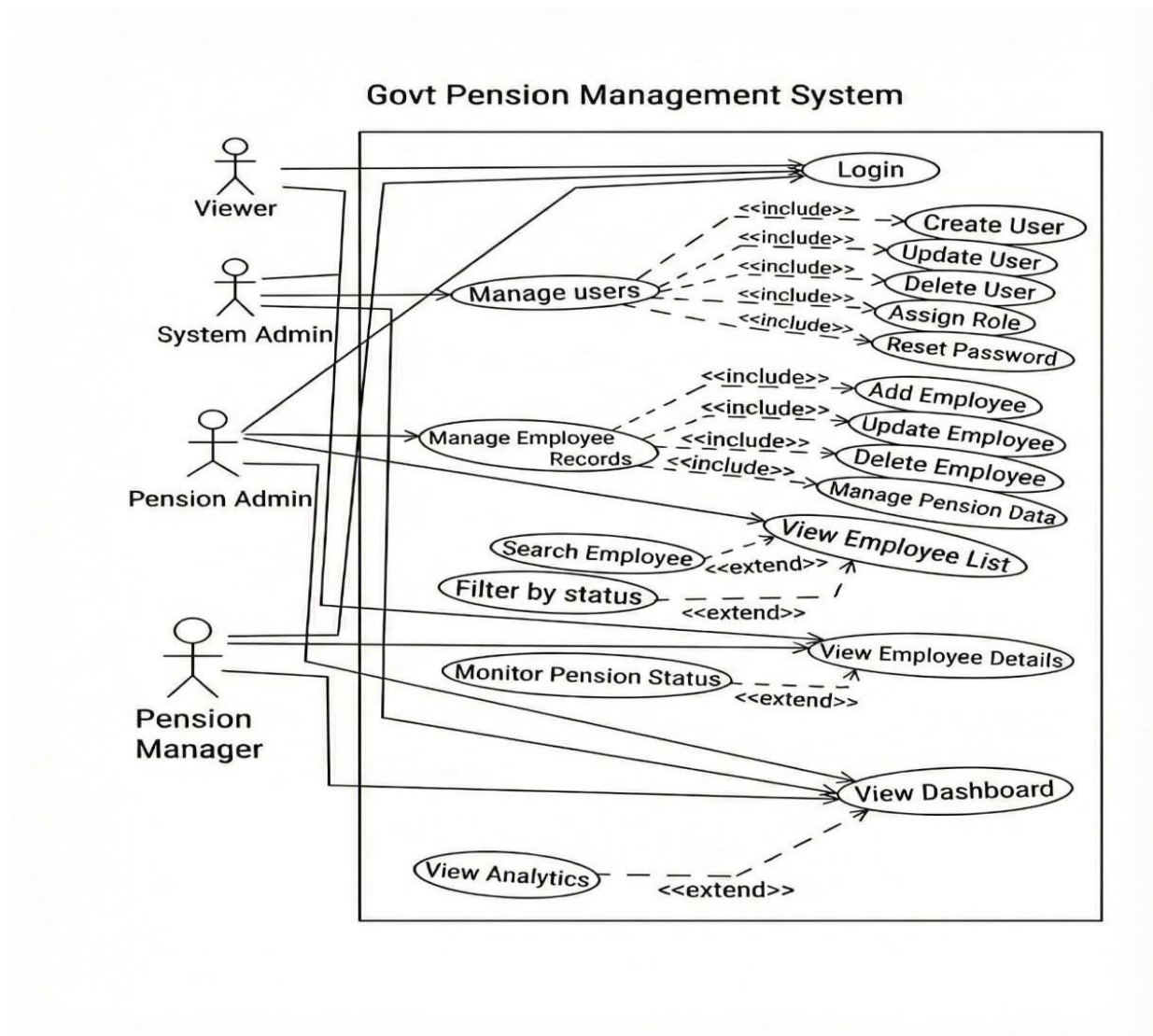


Fig-2 : Use Case Diagram

The Use Case Diagram of the Government Pension Management System (Govt. BD Pension) shows how different users interact with the system based on their roles. The system includes four main actors: System Admin, Pension Admin, Pension Manager, and Viewer. Each actor must log in using valid credentials before accessing the system, and all activities end with a logout, ensuring secure and controlled system usage.

The System Admin has full control over the system. This role is responsible for managing system users, assigning roles, resetting passwords, and accessing all

administrative dashboards. The Pension Admin handles core pension operations such

as managing employee records, updating pension-related information, viewing employee lists, and accessing dashboard analytics for decision-making.

The Pension Manager mainly performs monitoring and supervisory tasks. This user can view employee pension details, check pension status, and access system dashboards in read-only mode without modifying any data. The Viewer has the most restricted access and can only view basic employee information and employee lists.

The diagram uses «include» relationships to represent essential system functions such as login and core data access, which are required for all users. «extend» relationships are used to show optional or additional actions like searching, filtering, monitoring, and viewing analytics. Overall, the Use Case Diagram clearly demonstrates secure, role-based interaction and an organized workflow for managing government pension information.

4.2. Activity Diagram

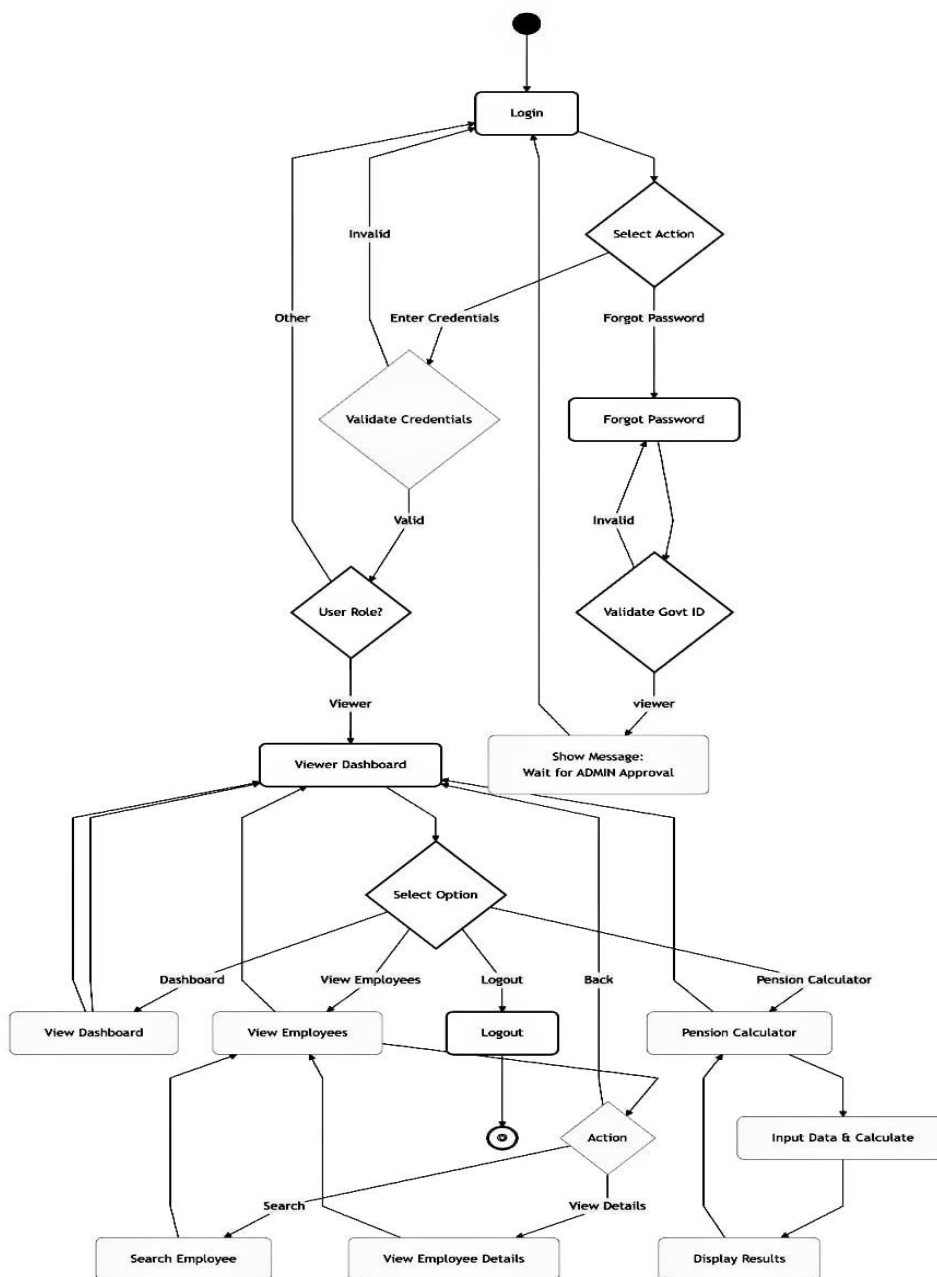


Fig-3 : System Admin Activity Diagram

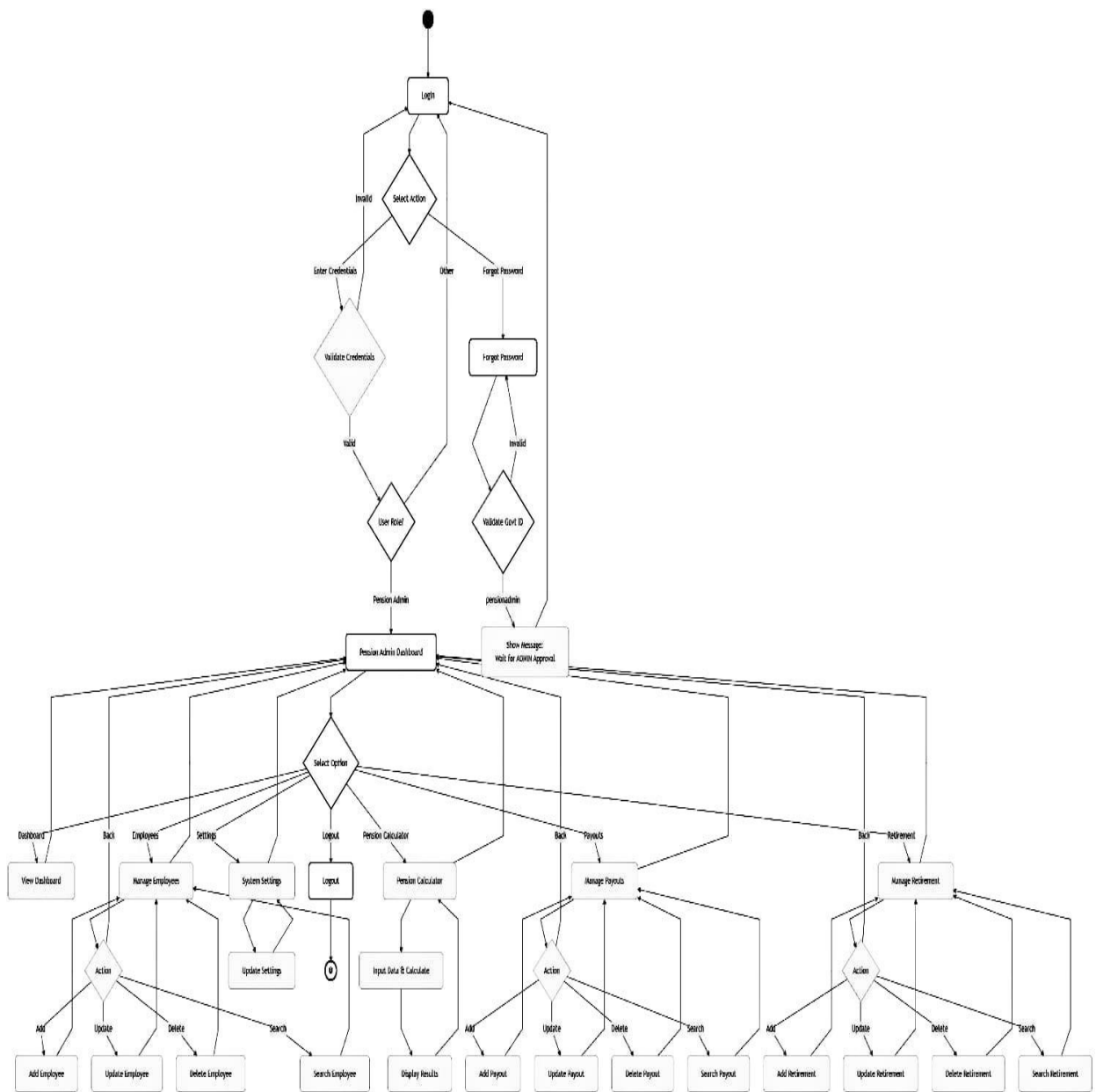


Fig-4 : Pension Admin Activity Diagram

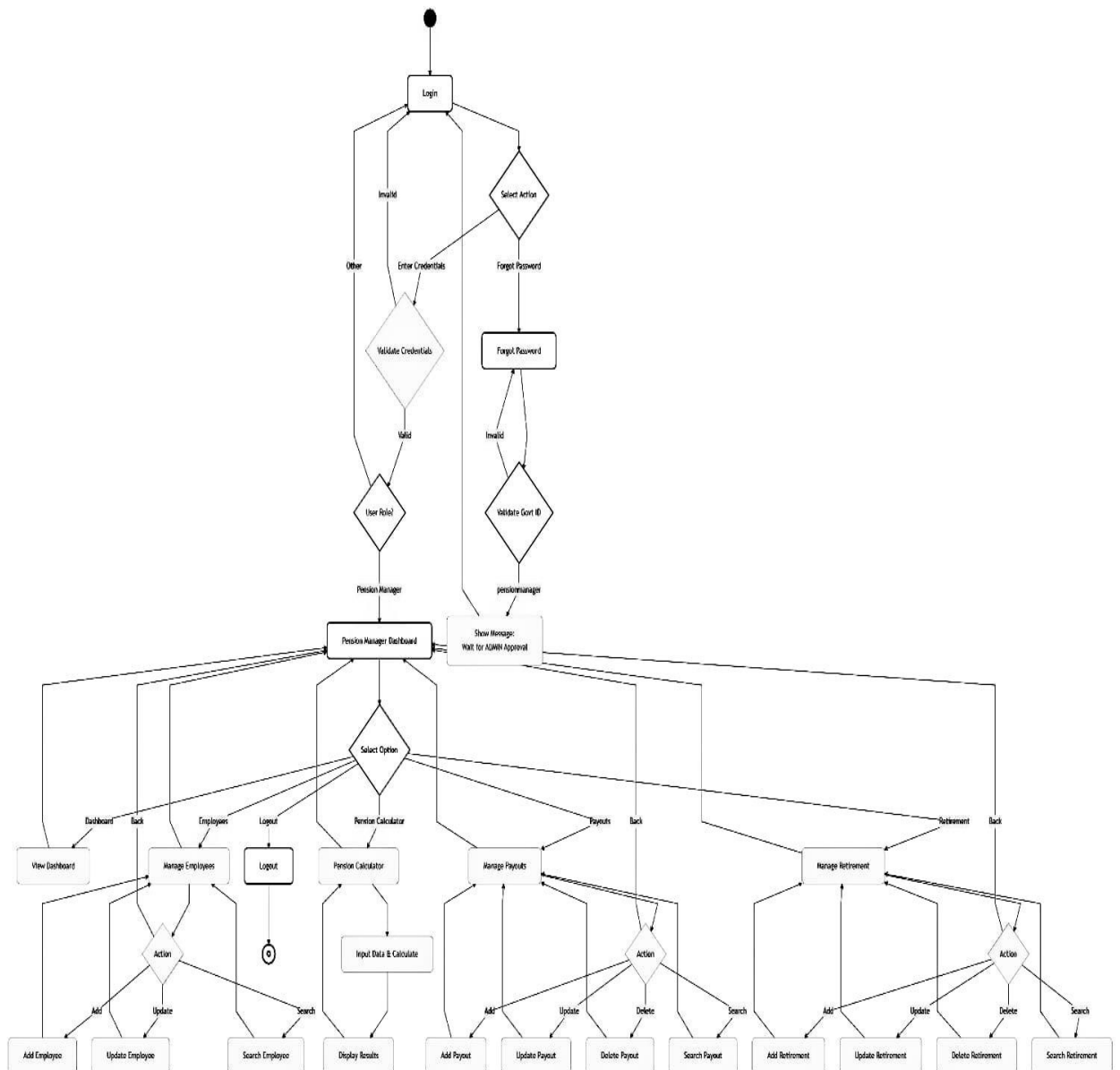


Fig-5 : Pension Manager Activity Diagram

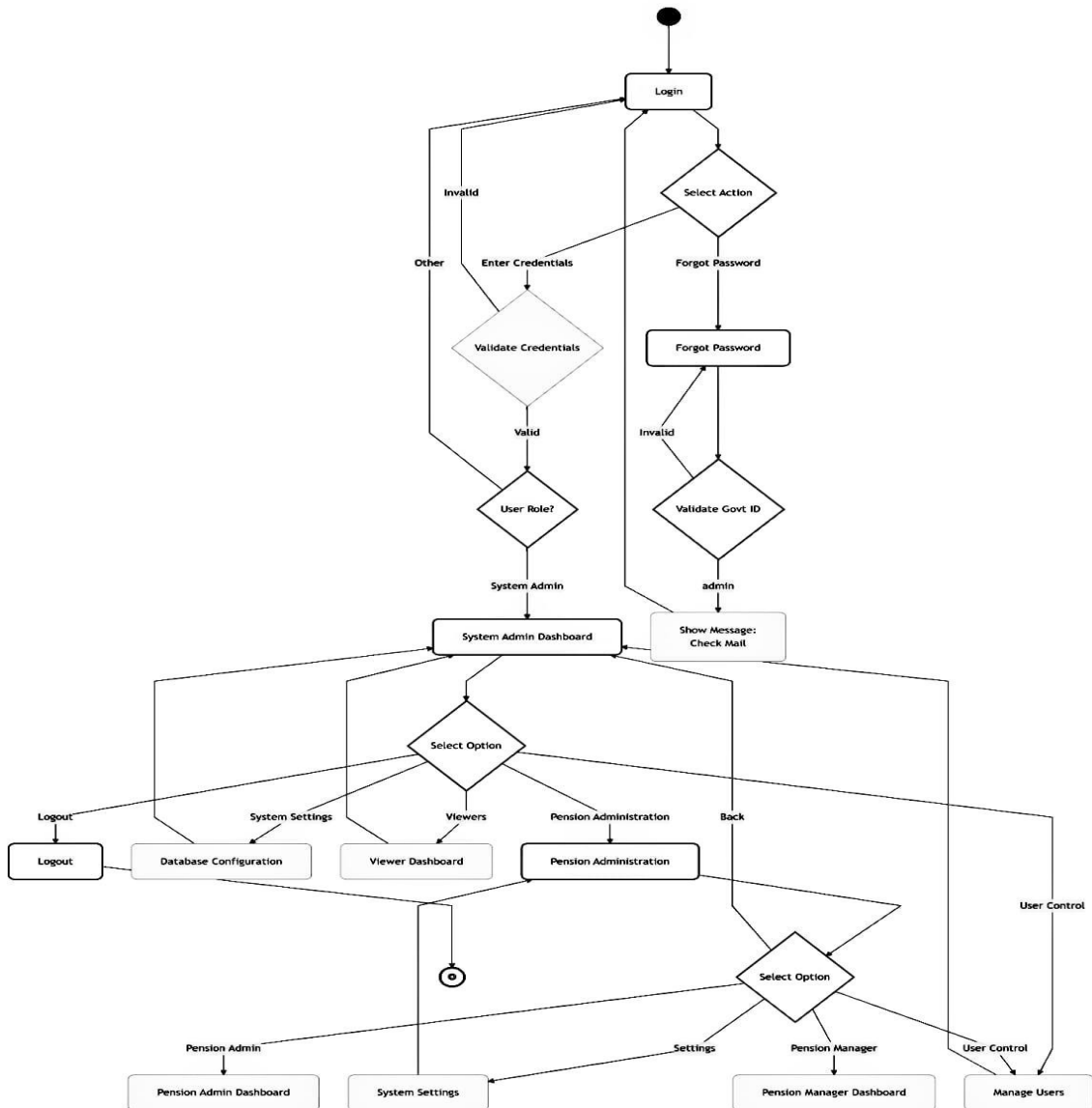


Fig-6 : Viewer Activity Diagram

The Activity Diagrams of the Govt. BD Pension Management System illustrate the overall workflow of the system based on different user roles. The system behavior is divided into four separate activity diagrams for System Admin, Pension Admin, Pension Manager, and Viewer, showing how each user interacts with the system after successful authentication.

The workflow begins with the Start node, where users access the login screen and enter their credentials. If the credentials are invalid, the system displays an error message and redirects the user back to the login page. After successful login, the system identifies the user role and redirects the user to the appropriate dashboard.

Each role follows a specific activity flow. The System Admin manages users, system settings, and pension administration modules. The Pension Admin handles employee management, pension calculation, payouts, and retirement records. The Pension Manager focuses on monitoring pension-related operations with limited administrative

control. The Viewer has read-only access to employee information, dashboards, and pension calculations.

The diagrams also include decision points to validate user actions and parallel activities using fork and join nodes, especially during pension calculation and retirement processing. These parallel flows improve efficiency by allowing multiple operations to occur at the same time.

All activities eventually lead to the Logout process, which safely ends the user session and marks the End of the workflow. Overall, the activity diagrams clearly represent role-based access control, validation, error handling, and the smooth operational flow of the pension management system.

All system activities eventually lead to the Logout process, which safely ends the user session and returns the system to the login screen. This diagram clearly demonstrates role-based access control, proper validation, and an organized operational flow of the pension management system.

5. CONCLUSION

Managing government pension data manually or through unstructured systems creates significant challenges, including data inconsistency, delayed processing, and lack of accountability. Such issues can negatively impact both administrative efficiency and employee trust.

The Government Pension Management System (Govt. BD Pension) successfully addresses these problems by offering a secure, structured, and role-based digital solution. Through proper implementation of C# GUI components, OOP principles, and database connectivity, the system ensures accurate data handling, efficient searching, and controlled access to sensitive information.

This project effectively demonstrates the practical use of object-oriented programming in solving a real-life problem. It fulfills academic requirements while maintaining scalability for future enhancements such as online payments, report generation, and web-based deployment. Overall, the system stands as a reliable, user-friendly, and professional pension management solution suitable for both academic evaluation and real-world application.